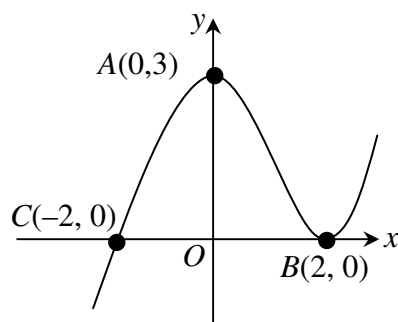


- 1 Describe a sequence of geometrical transformations that mapped the curve  $y^2 = x$  to the curve  $(y + 4)^2 = 2x + 7$ . [2]

2



The diagram above shows the curve  $y = f(x)$ . There is a maximum point at  $A(0, 3)$ , a minimum point at  $B(2, 0)$  and the curve also cuts the  $x$ -axis at the point  $C(-2, 0)$ . Sketch on separate diagrams, showing the corresponding points, if possible, the graphs of

- (i)  $y^2 = f(x)$ , [2]
  - (ii)  $y = f\left(\frac{x}{2}\right)$ , [2]
  - (iii)  $y = f'(x)$ , where  $f'(x)$  is the derivative of  $f(x)$ . [2]
- 3 Write down the first three terms in ascending powers of  $x$  of the expansion of  $(1 - x)^{-2}$ . State the coefficient of  $x^n$  and the range of values of  $x$  for which the expansion is valid. Hence, or otherwise, find the value of  $\sum_{n=1}^{\infty} \frac{n}{2^n}$ . [5]
- 4 It is given that  $y = \tan(e^x - 1)$ .
- (i) Prove that  $\frac{d^2 y}{dx^2} = \frac{dy}{dx} (1 + 2e^x y)$ . [2]
  - (ii) Find Maclaurin's series for  $y$  in ascending powers of  $x$ , up to and including the term in  $x^3$ . [3]
  - (iii) Deduce the first three terms of the Maclaurin's series for  $\frac{\sec^2(e^{-x} - 1)}{e^x}$ . [2]

- 5  $A_1, A_2, A_3, \dots, A_n$  are defined as follows :

$$A_1 = 2, \quad A_{n+1} = 2(A_1 + A_2 + \dots + A_n) \quad \text{for } n = 1, 2, 3, \dots$$

- (i) Evaluate the numerical values of  $A_2, A_3$  and  $A_4$ . [1]
- (ii) Prove that  $A_2, A_3, A_4, \dots, A_n$  form a geometric progression. [3]
- (iii) Find the sum of the  $n$  terms of  $A_1 + A_2 + A_3 + \dots$ . [3]

- 6** Three functions  $f$ ,  $g$  and  $h$  are defined as follows :

$$f : x \rightarrow (x-2)^2 - 3, \quad x \leq a$$

$$g : x \rightarrow \cos x, \quad 0 \leq x < \frac{\pi}{2}$$

$$h : x \rightarrow x^2, \quad x \leq 1$$

- (i) State the largest value of  $a$  for which the inverse function  $f^{-1}$  exists, and define  $f^{-1}$  for this case. [4]
- (ii) Sketch on the same the diagram, the graphs of  $y = f(x)$ ,  $y = f^{-1}(x)$  and  $y = ff^{-1}(x)$ . Hence, solve the equation  $f^{-1}(x) = f(x)$ , leaving your answer in exact form. [4]
- (iii) Find the domain of  $h'$  where  $h'$  is a one-one restricted function of  $h$  such that the composite function  $gh'$  exists and gives a maximal range. Define this composite function  $gh'$ . [4]

**7** (a) Find  $\int \frac{x \ln(4x^2 + 1)}{4x^2 + 1} dx$ . [2]

(b) (i) Use the substitution  $u = \sqrt{x}$ , show that  $\int_4^9 \frac{\sqrt{x}}{x-1} dx = \ln \frac{3}{2} + 2$ . [3]

(ii) Hence, find  $\int_4^9 \frac{\ln(x-1)}{\sqrt{x}} dx$  exactly. [3]

- 8** The curve  $C$  has equation  $y = \frac{x^2 + kx + 1}{2x + 3}$  where  $k$  is a positive constant.

- (i) Obtain the equations of the asymptotes of  $C$ . [3]
- (ii) Find the value of  $k$  for which the  $x$ -axis is a tangent to  $C$ . [3]
- (iii) Sketch  $C$  for the case  $k = 3$ . Hence, using graphical method, find the range of values of  $x$  that satisfy the inequalities  $|x^2 + 3x + 1| > |4x + 6|$ . [4]

- 9** Referred to the origin  $O$ , the position vectors of  $A$ ,  $B$  and  $C$  are  $\mathbf{a} = 12\mathbf{i} + 8\mathbf{j} + \mathbf{k}$ ,  $\mathbf{b} = 3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$  and  $\mathbf{c} = -3\mathbf{i} - 2\mathbf{j} + p\mathbf{k}$  respectively.

- (i) Find the value of  $p$  if  $A$ ,  $B$  and  $C$  are collinear. [2]
- (ii) If  $p$  is a variable, the locus of  $C$  will be a straight line. Find the vector equation of this straight line,  $L_1$ . [2]
- (iii) Given another line  $L_2 : \frac{x+4}{2} = -y+2 = z-1$ , find the acute angle between the lines  $L_1$  and  $L_2$ . [3]
- (iv) Find the shortest distance from  $A$  to the line  $L_2$ . [2]

- 10** Sketch the curve  $y = \frac{1}{\sqrt{1 + \cos x}}$  for  $0 \leq x < \pi$ . [1]

$R$  is the region enclosed by the curve  $y = \frac{1}{\sqrt{1 + \cos x}}$ , the  $y$ -axis and the line  $y = 1$ .

- (i) Calculate the area  $R$ . [2]  
 (ii) Find the exact value of the volume generated when  $R$  is rotated  $2\pi$  radians about the  $x$ -axis. [3]

- 11** The positive number  $x_n$  satisfy the relation

$$x_{n+1} = \frac{1}{2} \left( x_n + \frac{a}{x_n} \right), n = 1, 2, 3, \dots, \text{ and } x_1 = a > 0$$

- (i) Given that  $\lim_{n \rightarrow \infty} x_n = l$ , find the  $l$  in terms of  $a$ . [3]  
 (ii) Show that  $x_n \geq \sqrt{a}$  for  $n \geq 2$ . [2]  
 (iii) Show that  $x_n \geq x_{n+1}$  for  $n \geq 2$ . [3]

- 12** (a) Show that the differential equation  $3xy^2 \frac{dy}{dx} + y^3 - 2x = 0$  may be reduced by

means of the substitution  $u = xy^3$  to  $\frac{du}{dx} = 2x$ . Solve the differential equation,

expressing  $y$  in terms of  $x$ . Sketch the family of solution curves of the differential equation. [4]

- (b) A population of insects is allowed to grow in an experimental environment. The rate of birth of the insects is proportional to the number (in thousands) of insects,  $x$ , at time  $t$  days after the start of the experiment. The insects died at a constant rate of 0.3 per day. Given that initially  $x = 5$  and it is decreasing at a rate of 0.1.

Show that the differential equation relating  $x$  and  $t$  is  $\frac{dx}{dt} = 0.04x - 0.3$ . [2]

Hence, solve the differential equation, expressing  $x$  in terms of  $t$ . [4]

With the help of a graph, explain why it is possible for the insects to be extinct in the experimental environment after some time. Find the duration for this to happen, giving your answer to 1 decimal place. [2]

**13 (a)** Given that  $z = 1 - \sqrt{3}i$ . Find the value of the real number  $b$  such that

$$\arg(z^2 + bz) = \frac{-5\pi}{6}. \quad [3]$$

**(b)** Use de Moivre's Theorem to show that the equation  $z^n = (z + i)^n, n \in \mathbb{Z}^+$  can be expressed as  $z = \frac{i}{e^{\left(\frac{2k\pi}{n}\right)i} - 1}$ . [2]

Hence or otherwise, show that  $z = \frac{1}{2} \left( \cot \frac{k\pi}{n} - i \right)$ . [2]

Write down the values of  $k$  that gives all the roots of the equations. [1]

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